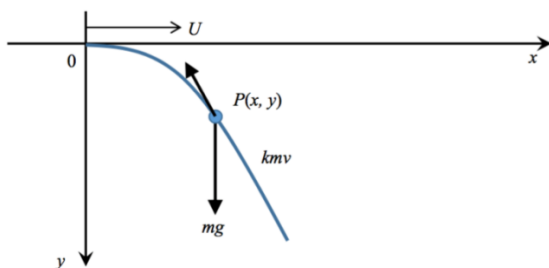
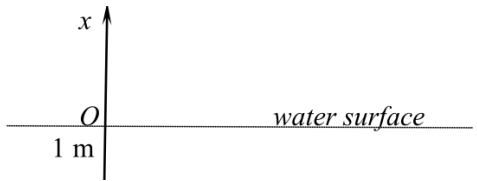


Mechanics WS 9

1. A bomb P of mass m is released from rest by a stealth bomber flying horizontally at a speed U . The bomb experiences the effect of gravity, and a resistance proportional to its velocity v in both the horizontal and vertical direction at any time t , where v is the speed of the bomber at time t . From the diagram, the equations of motion in the horizontal and the vertical directions are given respectively by $\ddot{x} = -k\dot{x}$ and $\ddot{y} = g - k\dot{y}$ [Do NOT prove this] where k is a constant and the acceleration due to gravity is g .



- Show that, at time t after the release, the bomb has travelled a horizontal distance $\frac{U(1-e^{-kt})}{k}$ metres
 - Show that, at time t after the release, the bomb is inclined at an angle $\tan^{-1} \left[\frac{g(e^{kt}-1)}{kU} \right]$ to the horizontal.
 - By considering the components of the velocity of the bomb, show that terminal velocity, V , of the bomb is $V = \frac{g}{k}$
 - Show that the least speed of the bomb is W where $W^2 = \frac{U^2 V^2}{U^2 + V^2}$ for $t \geq 0$.
2. A particle is moving in a straight line. After time t seconds it has displacement x metres from a fixed point O on the line, velocity $v = \frac{1-x^2}{2} \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. Initially the particle is at O .
- Find an expression for a in terms of x .
 - Show that $\frac{2}{1-x^2} = \frac{1}{1+x} + \frac{1}{1-x}$ and hence find an expression for x in terms of t .
 - Describe the motion of the particle, explaining whether it moves to the left or right of O , whether it slows down or speeds up, and where its limiting position is.
3. A solid of unit mass is dropped under gravity from rest at a height of h metres. Air resistance is proportional to the speed v of the mass, acceleration under gravity is g .
- Using k as the constant of proportionality, show that the acceleration is given by $\frac{d^2x}{dt^2} = g - kv$
 - Show that the velocity v of the solid after t seconds is given by $v = \frac{g}{k}(1 - e^{-kt})$
 - Using the fact that $\frac{d^2x}{dt^2} = v \frac{dv}{dx}$ show that $x = \frac{g}{k^2} \left[\ln \frac{g}{g-kv} - \frac{kv}{g} \right]$
4. A particle is projected from the origin with speed V at an angle α to the horizontal. The particle is subject to both gravity and an air resistance proportional to its velocity, so that its respective horizontal and vertical components of acceleration while it is rising are given by $\ddot{x} = -k\dot{x}$, $\ddot{y} = -g - k\dot{y}$
- Show that
 - $\dot{x} = (V \cos \alpha)e^{-kt}$
 - $\dot{y} = \left(\frac{g}{k} + V \sin \alpha \right) e^{-kt} - \frac{g}{k}$
 - Hence show that $x = \frac{V \cos \alpha}{k}(1 - e^{-kt})$
 - When the particle has reached its greatest height, show that it has travelled a horizontal distance of $\frac{V^2 \sin 2\alpha}{2(g + V k \sin \alpha)}$
5. A particle mass m moves in a horizontal straight line. The particle is resisted by a constant force mk and a variable force mv^2 , where k is a positive constant and v is the speed. When $t = 0$, $v = u$. Find the distance travelled and the time taken as the particle is brought to rest.
6. The depth of water in a harbour is 7.2 m at low water and 13.6m at high water. On Monday, low water is at 2.05 pm and high water at 8.20 pm. The captain of a ship drawing 12.3 m of water wants to leave harbour as early on Monday afternoon as he can. Find

- a. Between what times he can leave on Monday
- b. His earliest leaving time on Wednesday if he fails to leave on Monday or Tuesday
7. A particle of mass 0.5 kg is released from rest and moves vertically downward under gravity in a medium which exerts a resistance to the motion of $\frac{1}{10}v^2$. At time t after release it has fallen a distance x and has velocity v . Taking $g = 10 \text{ ms}^{-2}$. Show that $v^2 = 50(1 - e^{-0.4x})$ and $\ddot{x} = 10e^{-0.4x}$.
8. A ping pong ball is initially placed 1 metre beneath the surface of the water, as shown in the diagram. The ping pong ball is released in the water with an acceleration of $\ddot{x} \text{ m/s}^2$, where $\ddot{x} = -625x$ and where x metres is the displacement of the motion measured from the water surface.
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- a. Is the motion of the ping pong ball only SHM? Give reasons.
- b. Prove that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$
- c. Find the expression for the ball's velocity $v \text{ m/s}$ when it is in the water.
- d. Find the velocity of the ball at the water's surface
- e. Assuming there is no air resistance and the acceleration due to gravity is 10 m/s^2 , derive an expression for the displacement in air in terms of v
- f. Find the maximum height that the ball reaches above the surface of the water.
9. A forklift is driving down a warehouse aisle. The acceleration of the forklift is given by the equation $\ddot{x} = -\frac{1}{2}\mu^2 e^{-x}$, where x is the displacement from the origin and μ is the initial velocity at the origin
- a. Show that $v^2 = 4e^{-x}$ if $\mu = 2 \text{ m/s}$
- b. Explain why $v > 0$
- c. Find an equation for x in terms of t
- d. Describe the motion of the particle as $t \rightarrow \infty$
10. The acceleration $a \text{ cm/s}^2$ of a particle P is given by $a = 18x(x^2 + 1)$, where $x \text{ cm}$ is the displacement of P at t seconds. Initially P starts from the origin with velocity 3 cm/s .
- a. Show that $v = 3(x^2 + 1)$,
- b. And show that $x = \tan 3t$
- c. Determine its velocity and displacement after $\frac{\pi}{12} \text{ s}$
11. A particle moves in a straight line along the x axis so that its acceleration is given by: $\ddot{x} = \frac{1}{4+x^2}$ where x is the displacement from the origin. Initially the particle is at rest at the origin.
- a. Find v^2 as a function of x
- b. Explain why v is always positive for $t > 0$.
12. A particle moves in a straight line so that after t seconds its velocity is $v \text{ m/s}$ and its displacement is x .
- a. Given that $\frac{d^2x}{dt^2} = 10x - 2x^3$ and that $v = 0$ when $x = -1$, find v in terms of x .
- b. Explain why the motion cannot exist between $x = -1$ and $x = 1$

Answers:

2. (a) $\frac{x^3-x}{2}$; (b) $x = \frac{e^t-1}{e^t+1}\left(\frac{1-e^{-t}}{1+e^{-t}}\right)$; (c) moves to the right, slowing down. Limiting position is $x = 1$.
5. $\frac{1}{2}\ln\left(1 + \frac{1}{k}u^2\right)$; $\frac{1}{\sqrt{k}}\tan^{-1}\frac{u}{\sqrt{k}}$
- 6.
- 7.
8. $v = 25\sqrt{1-x^2}$, $v = 25$, $x = \frac{625-v^2}{20}$, 31.25 m
9. $x = 2\ln(t+1)$, particle stops
10. c. $x = 1$, $v = 6$
11. a. $v^2 = \tan^{-1}\left(\frac{x}{2}\right)$; b. $\ddot{x} = \frac{1}{4+x^2} > 0$ for all values of x , Since particle begins at rest, positive acceleration means the particle moves in the positive direction and will not slow down unless acceleration becomes negative.
12. a. $v = \pm\sqrt{10x-x^4-9}$; b. $v^2 < 0$ for $-1 < x < 1$